

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: ENGINEERING PHYSICS

Branch : ALL

Semester :II

Subject Code & Name: BTBS202P

Max Marks: 60

Date: 02/07/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Piezoelectric method is used when frequencies needed are a. 200 KHZ b. 50 KHZ c. 100 KHZ d. 500 KHZ	REMEMBER (CO1)	1
2	The oscillatory motion is said to be over damped if ----- a. $b^2 > \omega^2$ b. $b^2 < \omega^2$ c. $b^2 = \omega^2$ d. None of these	REMEMBER (CO1)	1
3	In Newton's ring experiment the center appears to be a. Bright b. Dark c. Both a and b d. None of these	REMEMBER (CO2)	1
4	The substance which rotates the plane of polarization to right is called as a. Dextrorotatory b. Levorotatory c. Oscillatory d. None of these	REMEMBER (CO2)	1
5	Wavelength of Ruby laser is a. $6943 \text{ \AA}$ b. $6428 \text{ \AA}$ c. $6328 \text{ \AA}$ d. $6398 \text{ \AA}$	REMEMBER (CO2)	1
6	Optical fiber operates on the principle of a. Total internal reflection b. Total external reflection c. Total internal refraction d. Total external refraction	REMEMBER (CO2)	1
7	Dual nature [particle and wave] of matter was proposed by a. de Broglie b. Planck c. Newton d. Heisenberg	REMEMBER (CO3)	1
8	The mixture filled in G.M. tube is a. $\text{Ar} + \text{C}_2\text{H}_5\text{OH}$ b. $\text{Ne} + \text{C}_2\text{H}_5\text{OH}$ c. $\text{Ar} + \text{CH}_3\text{OH}$ d. $\text{Ne} + \text{CH}_3\text{OH}$	REMEMBER (CO3)	1
9	Number of atoms per unit cell for Face centered Cubic structure is a. 2 b. 6 c. 4 d. 12	REMEMBER (CO4)	1
10	Coordination Number for Body centered Cubic structure is a. 6 b. 8 c. 12 d. 10	REMEMBER (CO4)	1
11	For superconductor susceptibility is a. 0 b. 1 c. -1 d. 2	REMEMBER (CO5)	1
12	Hall effect is associated with	REMEMBER	1

	a. Conductors	b. Semiconductors	c. Thermistors	d. Both a & b	(CO5)	
Q. 2	Solve the following.					12
A)	Define damped vibrations. Derive an expression for damped vibrations.				REMEMBER & UNDERSTAND (CO1)	6
B)	State Piezoelectric effect. Explain Piezoelectric method for the production of ultrasonic waves.				REMEMBER & UNDERSTAND (CO1)	6
Q.3	Solve the following.					12
A)	Explain the construction & working of Helium Neon laser.				UNDERSTAND (CO2)	6
B)	Derive an expression for numerical aperture of optical fiber.				UNDERSTAND (CO2)	6
Q. 4	Solve Any Two of the following.					12
A)	Explain the construction & working of Bainbridge mass spectrograph with labeled diagram.				UNDERSTAND (CO3)	6
B)	Derive Schrodinger's time dependent wave equation.				UNDERSTAND (CO3)	6
C)	Explain construction & working of Geiger Muller counter with neat labelled diagram.				UNDERSTAND (CO3)	6
Q.5	Solve Any Two of the following.					12
A)	Calculate atomic packing factor for Simple cubic, body centered cubic & face centered cubic structure.				UNDERSTAND (CO4)	6
B)	Explain continuous & characteristics X-ray spectra.				UNDERSTAND (CO4)	6
C)	Derive the relation between lattice parameter 'a' and crystal density 'ρ' Nickel is characterized by FCC lattice. The edge of the unit cell is $3.52 \text{ \AA}$. The atomic weight of nickel is 56.71. Determine the density of the metal. ($N = 6.02 \times 10^{23} \text{ mole}^{-1}$)				UNDERSTAND (CO4)	6
Q. 6	Solve Any Two of the following.					12
A)	Explain B-H curve with diagram. Hence define coercivity and retentivity				UNDERSTAND (CO5)	6
B)	State & explain Meissner effect for superconductors.				UNDERSTAND (CO5)	6
C)	Derive an expression for conductivity of intrinsic and extrinsic semiconductors.				UNDERSTAND (CO5)	6
	*** End ***					